

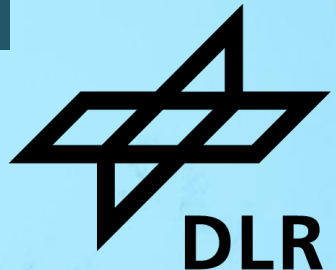
AN INTEGRATED METAHEURISTIC APPROACH FOR GENERATING COMBINED ROAD-RAIL TRANSPORT CHAINS

Theresa-Maria Mersini ^a, Ibraheem O. Adeniran ^a, Marcel Schindler ^b, Dr. Alessandro Tarantola ^b, Dr. Carina Kehrt ^a

^a Institute for Transport Research, ^b Institute of Quantum Technologies

MATSim User Meeting 2026

Paris, September 28, 2026



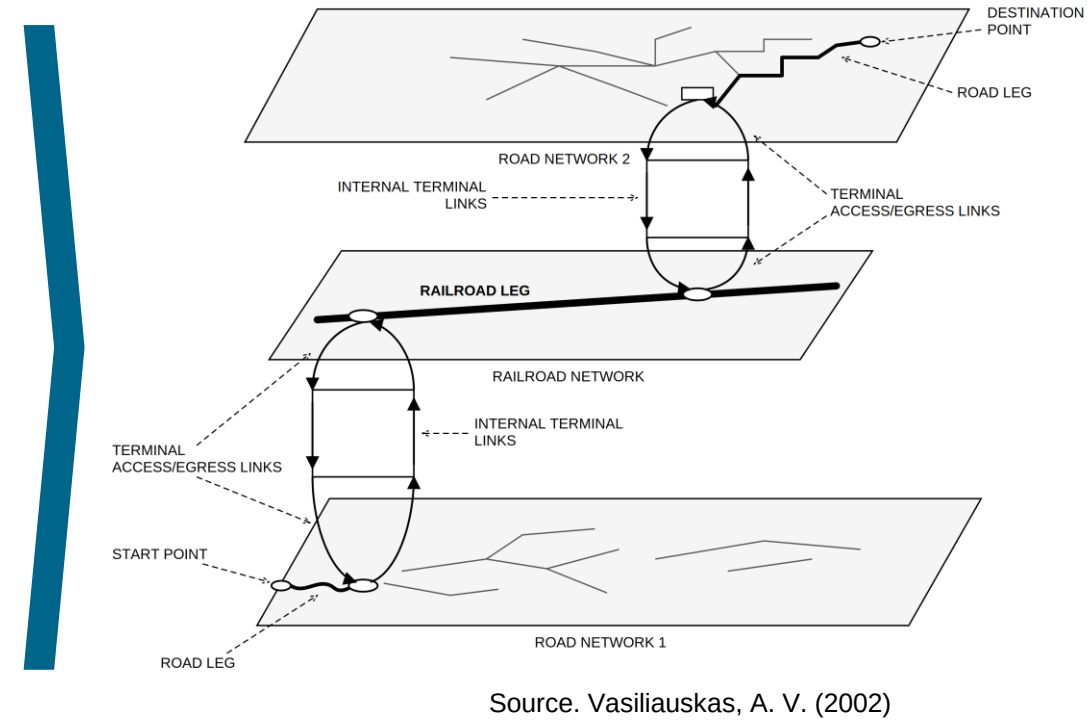
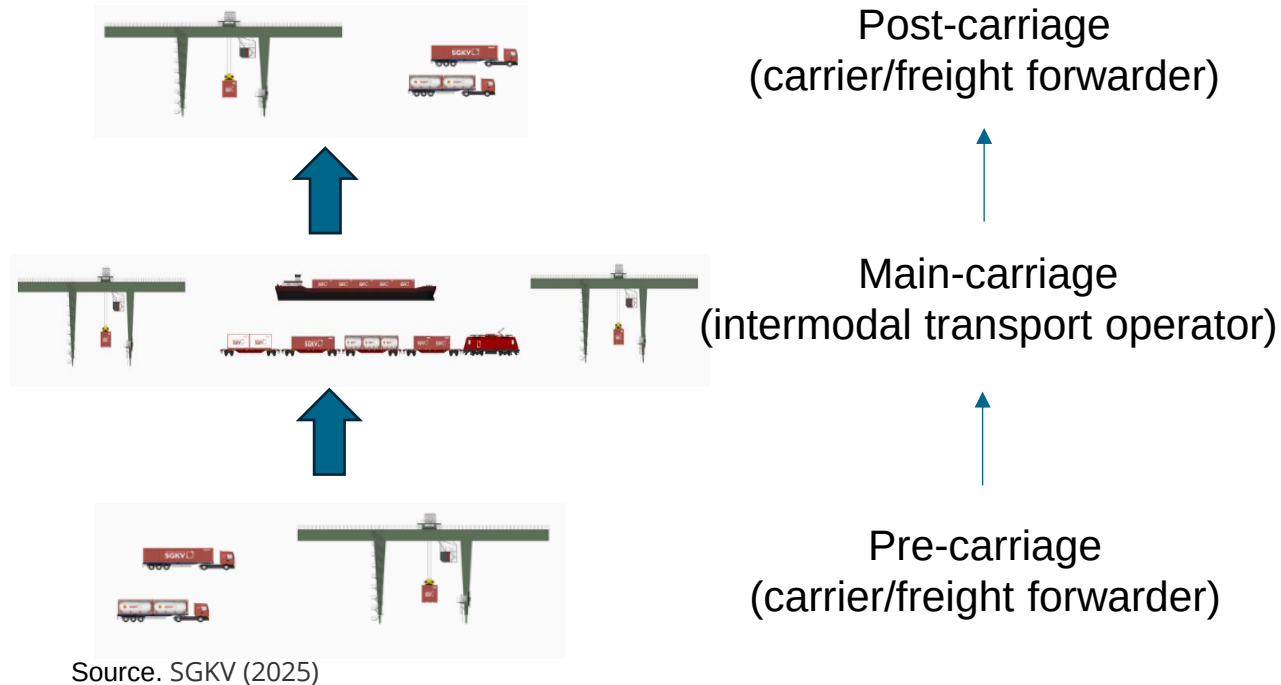
Agenda



- Motivation and Background
- Methodological Procedure and Methods used
- Optimisation-Simulation Workflow
 - Chain Construction
 - Plan Generation
 - Plan Execution using MATSim
- Conclusion and Outlook

Motivation and Background

Research Objective: Operational optimisation of intermodal transport chains within a nationwide network



The integration of pre-carriage, main carriage and post-carriage enable the analysis of complete **intermodal transport chains**

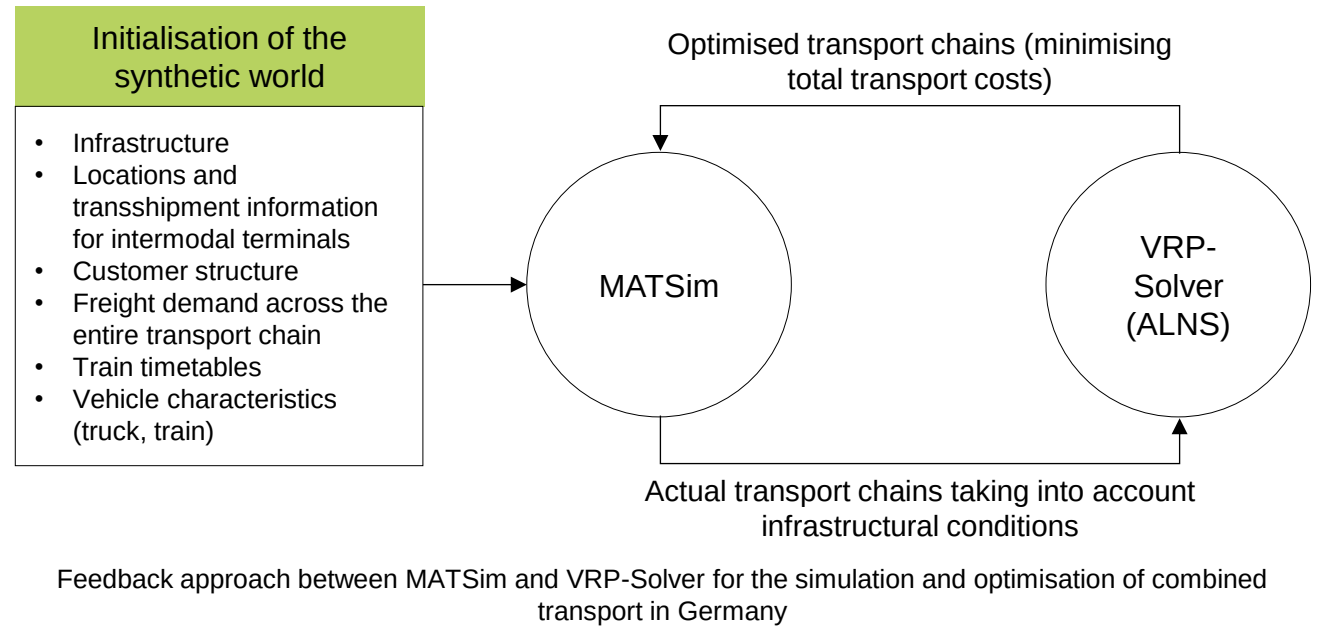
Methodological Procedure and Methods used

Problem formulation

- The intermodal coordination problem formulated as a **dynamic Multi-Commodity Network Flow Problem** with **temporal integration of different modes of transportation**

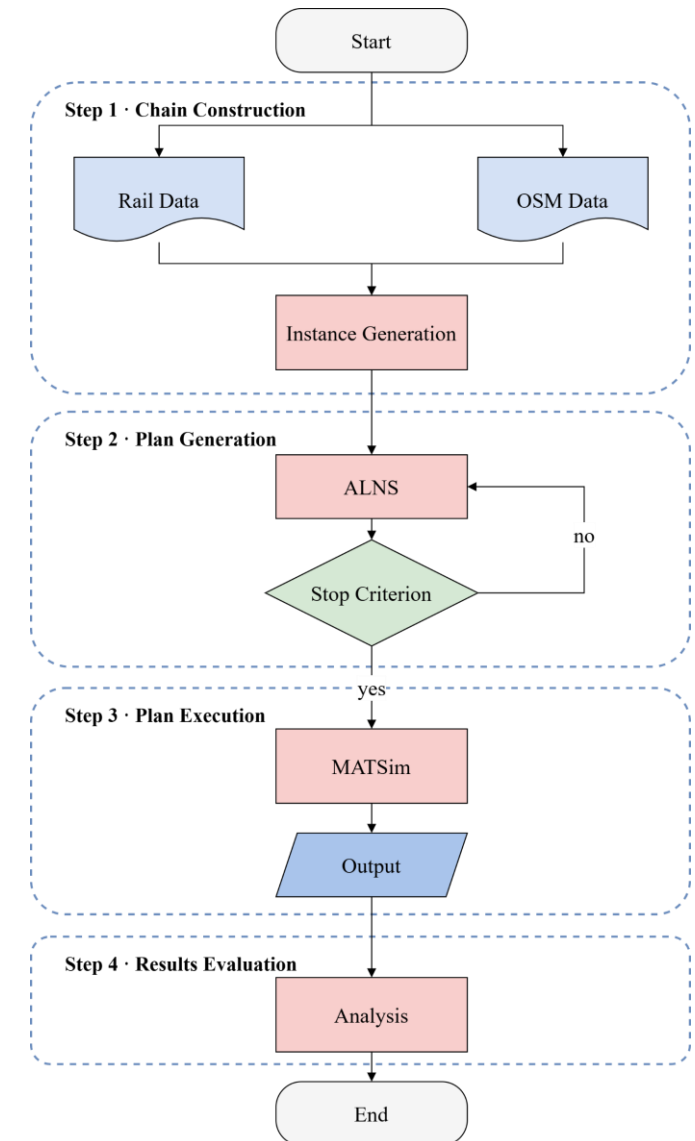
Methodological Approach

- Approximation of optimal solutions with heuristics, i.e. Adaptive Large Neighborhood Search (ALNS)
- Linking the results with a simulation environment for real world scenarios
- Case study:** German intermodal transport system



Metaheuristic-Simulation Workflow

- A workflow for evaluating intermodal road-rail transport plans from loading-unit data
- Rail and OSM data are integrated to generate instances of complete intermodal transport chains.
- The plan generation step is modular, in this case we are using an ALNS.
- The plans will be executed using MATSim.

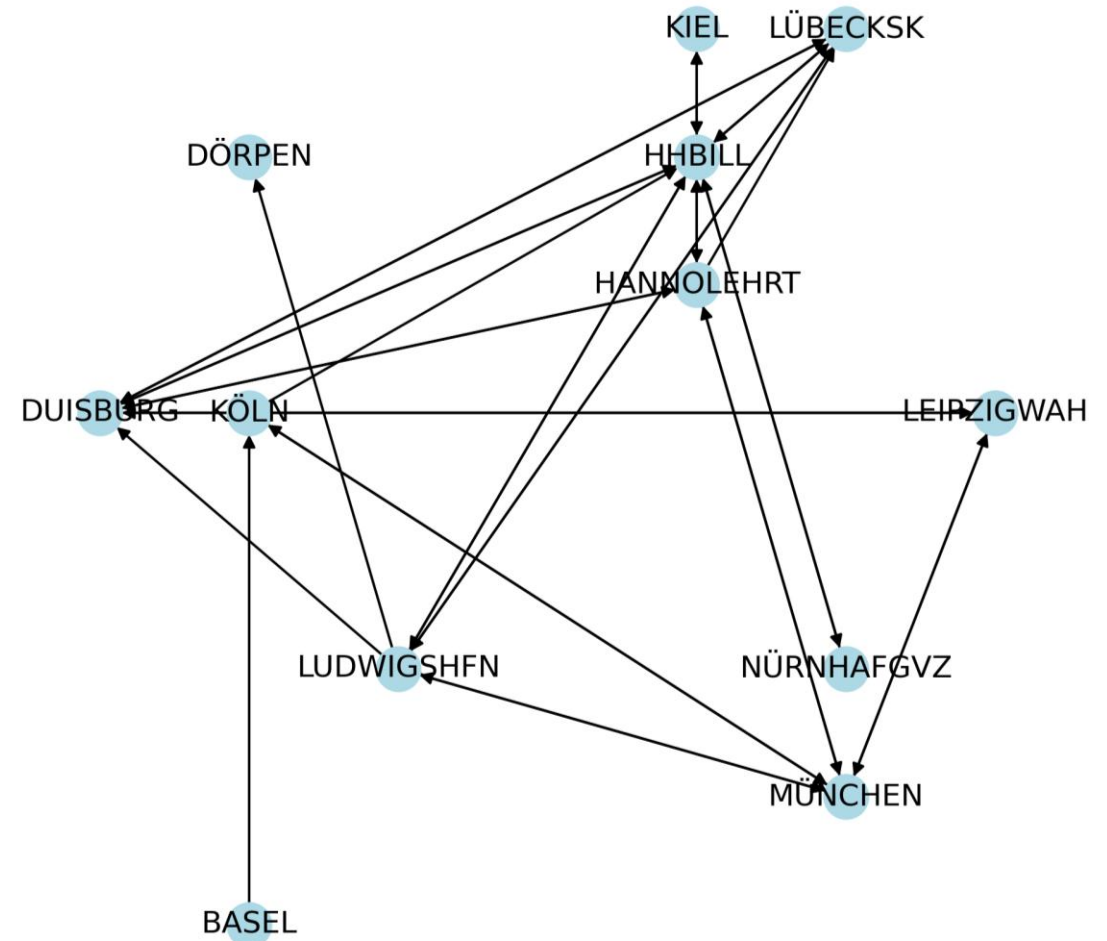


Intermodal Transport Data

- Historical transport data provided by a intermodal service provider
- Contains only rail leg data for loading units transported between intermodal terminals.

```
{  
  'origin_terminal': 'DUISBURG',  
  'destination_terminal': 'LÜBECKSK',  
  'origin': 'null',  
  'destination': 'null',  
  'start_date_range': '2025-05-19',  
  'latest_arrival': '2025-05-25',  
  'weight_t': 35.216,  
  'length_m': np.float64(13.6)  
}
```

Not included

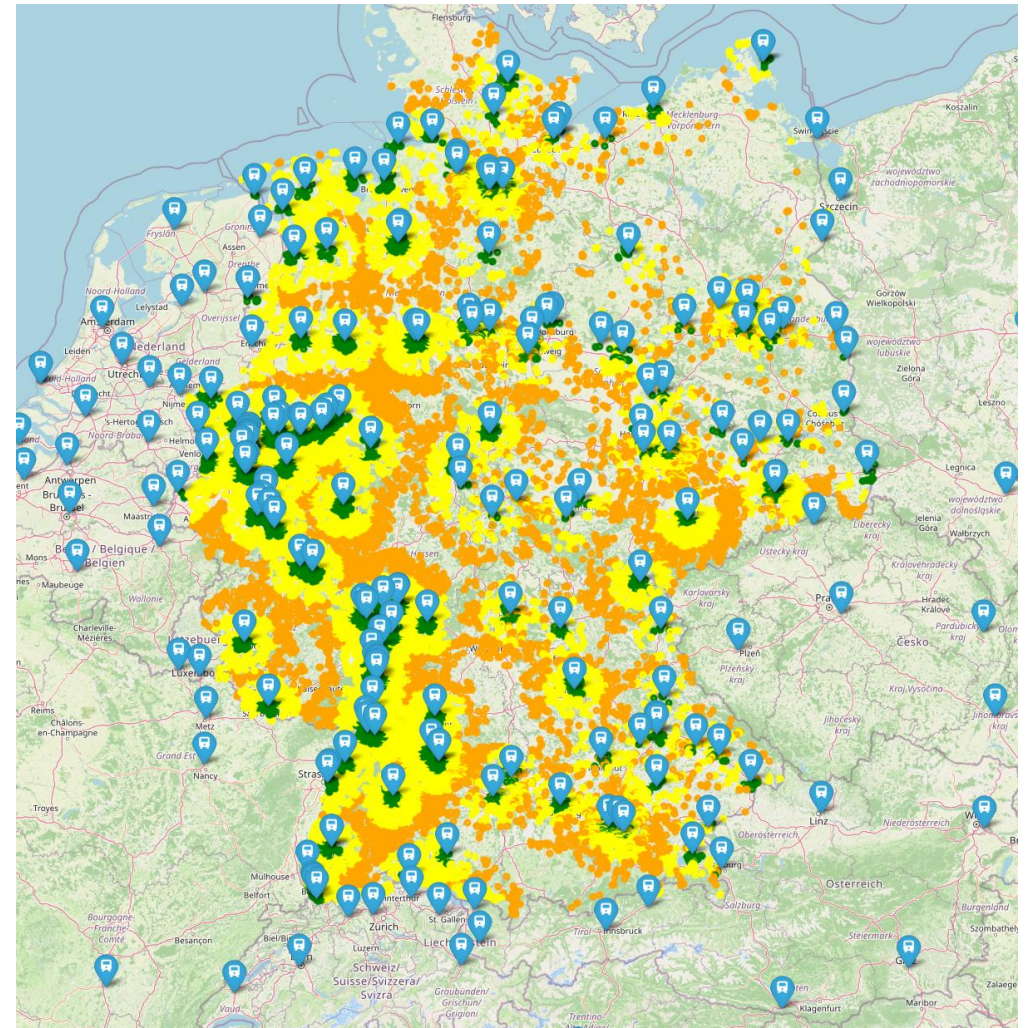


Intermodal Transport Data

OSM Extraction

- Extraction of synthetic origins and destinations using QuackOSM
- Query retrieves metadata for industrial facilities within a 50 km radius of intermodal terminals

```
OSM_TAGS = {  
  "industrial": [  
    "warehouse", "distribution", "logistics", "factory",  
    "brewery", "food", "chemical", "plastic_manufacturing",  
    "plastic_processing", "metal_processing",  
    "automotive_parts", "electronics", "textile", "depot"],  
  
  "building": ["warehouse", "industrial", "factory"],  
}
```

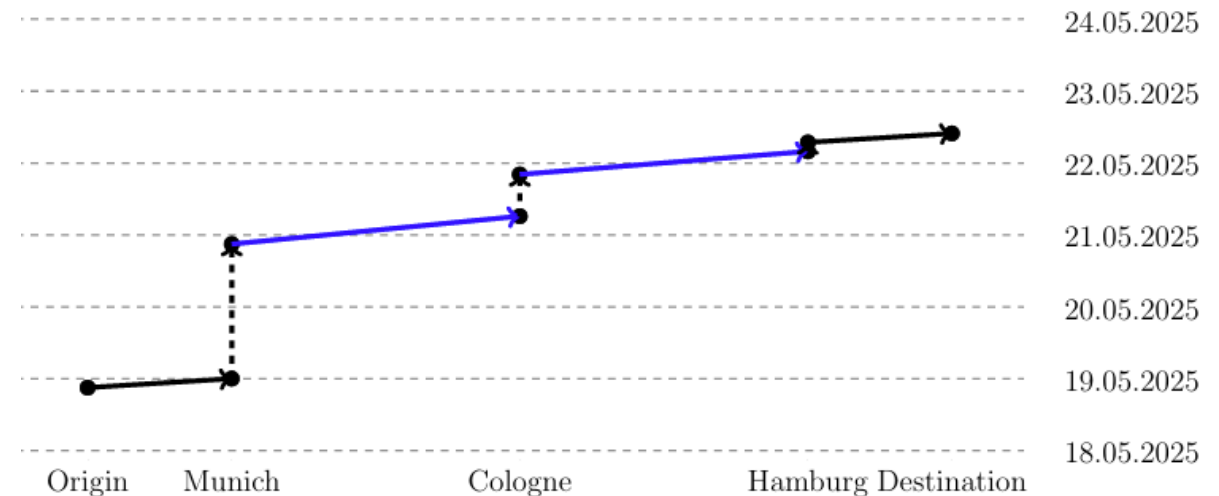


Creating Plans for each Loading Unit



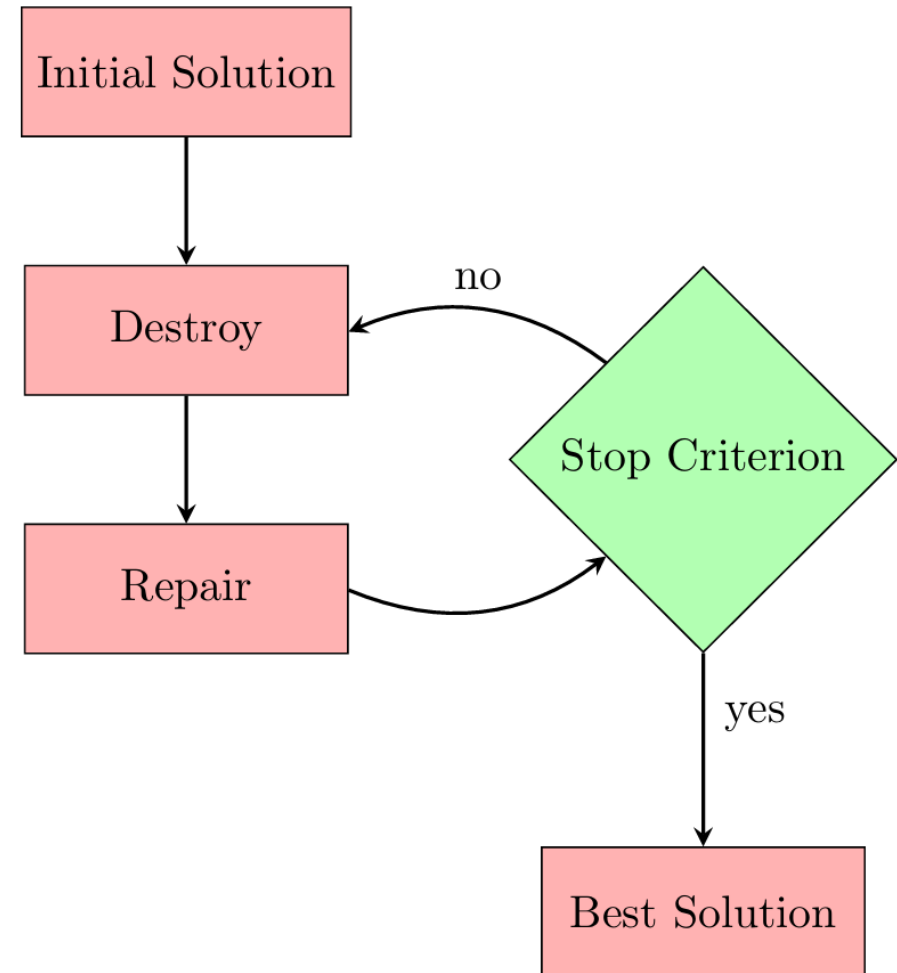
- Find the shortest path in a time dependent expansion for each start date
- Create an activity-trip chain from these paths
- Score the plans based on cost of road and rail transport and transshipment per hour at the terminal
- Create a population of loading units
- Use the set of rail services for capacity restriction in the ALNS

```
{  
  'origin_terminal': 'MUNICH',  
  'destination_terminal': 'HAMBURG',  
  'origin': 'null',  
  'destination': 'null',  
  'start_date_range': ['2025-05-19', '2025-05-20', ...],  
  'latest_arrival': '2025-05-25',  
  'weight_t': 35.216,  
  'length_m': np.float64(13.6)  
}
```



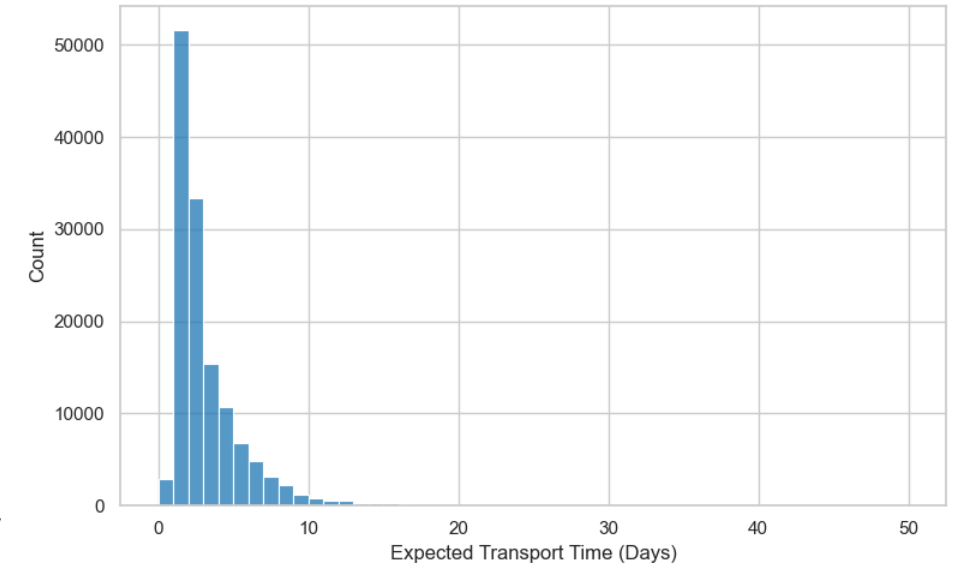
Plan Selection using an ALNS

- Create an initial feasible solution by randomly choosing a plan from the memory of each loading unit
- Destroy the current state by randomly removing 50% of the currently routed loading units
- Repair the solution by attempting to insert the plan with the lowest possible score for each unassigned loading unit, that does not violate the capacities
- Repeat until the stop criterion is met



Executing the Best Solution in MATSim

- Development of the MATSim simulation model is ongoing
- The model is based on SBB's MATSim intermodal freight scenario
- Each plan is simulated once, without iterative replanning, because optimisation is performed externally
- This approach enables the evaluation of plans generated by various optimisation methods
- Subsequent work will include simulating plans over a multi-day horizon



Conclusion & Outlook

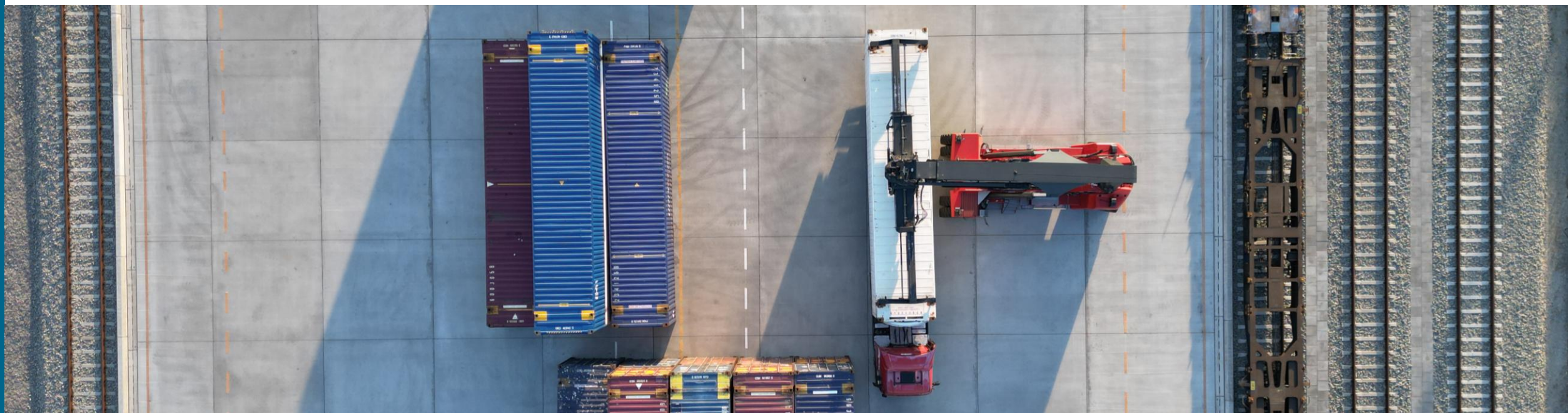


- The ALNS metaheuristic is modular, other methods can be used in its place.
- This approach enables testing loading unit plan choices against the rail services capacities
- Rerouting and the generation of new plans is currently not implemented but possible.
- We aim to use this approach for sensitivity testing and scenario analysis.

Thank you very much for your Attention!

Theresa-Maria Mersini

Contact: theresa.mersini@dlr.de



Topic: An Integrated Metaheuristic approach for generating combined road-rail transport chains

Date: 2026-09-28

Author: Theresa-Maria Mersini, Ibraheem O. Adeniran, Marcel Schindler, Alessandro Tarantola, Carina Kehrt

Institute: Institute of Transport Research, German Aerospace Center

Image sources: All images “DLR (CC BY-NC-ND 3.0)” unless otherwise stated